

HyperBones: Realtime Bone-driven Neural Garment Simulation with Hypernetwork Conditioning

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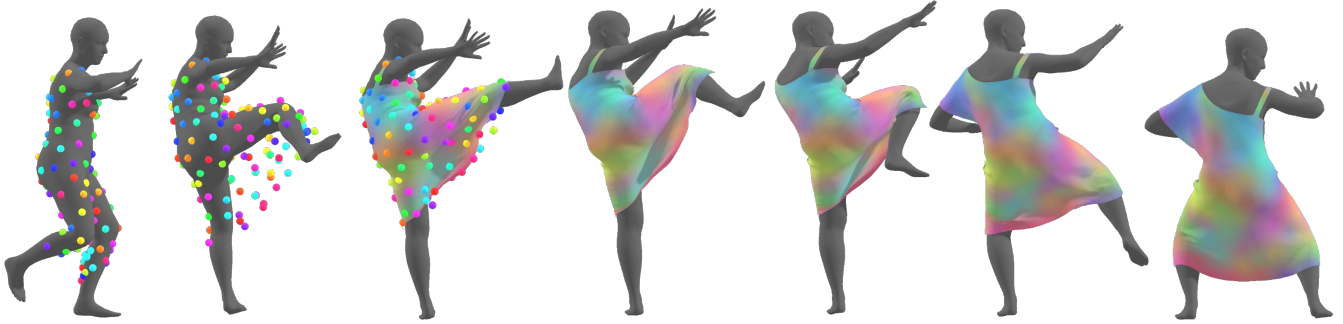


Fig. 1. We propose an efficient neural garment simulation method, which handles garments across a range of motions and body shapes for a fixed set of garments. We demonstrate 25 $\times$ performance improvement over state-of-the-art methods with comparable visual quality.

Recent advances in garment simulation have brought high-quality results closer to real-time performance. Physics-based simulators can produce accurate motion, but remain too computationally expensive for interactive applications. In contrast, linear blend skinning is efficient, but cannot capture the complex dynamics of loose-fitting garments, often leading to unrealistic motion and visual artifacts. Neural methods offer a promising alternative, yet they still struggle to animate loose clothing plausibly under strict run-time constraints. We present a fast and physically plausible approach for dynamic garment simulation. Our method trains a reduced-space neural dynamics simulator composed of independent coarse- and fine-level components. At the coarse level, the garment is driven by a set of virtual bones integrated with a lightweight neural network. Fine-scale wrinkle details are then recovered using a trained convolutional neural map. By decoupling identity-specific computation from real-time neural integration, our architecture maintains high performance while supporting diverse body shapes and motions. We further introduce an effective physics-supervision scheme that enables accurate results without relying on an external simulator. Experiments show that our method produces physically plausible garment dynamics, generalizes across a range of motions and body shapes, and supports a fixed set of garments. Our simulator runs at 300+ FPS on a commodity GPU, making it suitable for real-time applications.

1 Introduction

Character animation has been a central challenge in graphics and vision research, driving innovation across industries from video games and visual effects to emerging virtual and mixed reality experiences [WSS⁺25]. Despite significant progress, achieving high-quality animation, especially for clothing, remains difficult. This is largely due to the limited compute budgets available on most devices.

Traditional physics-based simulation methods, such as finite element and position-based techniques, are capable of producing realistic cloth motion. However, they suffer from critical drawbacks: high computational cost, lack of robustness, and the need for expert-driven parameter tuning. These factors make real-time performance at scale unattainable on anything but high-end hardware, and full automation for data generation impractical.

Recent learning-based approaches have sought to bridge the gap between realism and efficiency, but they often require substantial compute and memory resources [GBH23], or struggle to generalize across diverse body shapes, poses, and garment types [SOC22]. Fast skinning methods such as linear blend skinning (LBS) offer real-time performance at the expense of physical plausibility, frequently introducing artifacts that break visual immersion.

We address this limited expressiveness by introducing a set of *virtual bones* [PMJ⁺22] placed on the garment surface, which act as an intermediate, coarse level control structure between the articulated skeleton and the garment vertices. To learn physically accurate transformations of these virtual bones, we propose a self-supervised framework that does not depend on precomputed simulation data and demonstrates robust generalization across diverse body motion sequences. Our approach is motivated by the insight that worn garments exhibit different modes of deformation when observed from afar and up-close. From a distance, garment (especially loose garment) motion can be modeled by coarse level affine transforms defining its position and orientation in space. Wrinkle-level detail, on the other hand, stays close to the surface of the garment, and so is best modeled in UV space. This observation allows us to substantially reduce the runtime computational cost by decoupling coarse- and fine-level deformation into two stages.

In the first stage, the pose-dependent network takes as input a short history of virtual bone positions and velocities over consecutive frames and predicts corrected virtual bone transforms, which are applied via LBS to produce a coarse, globally coherent garment mesh. In the second stage, a convolutional network operating in UV space predicts per-vertex residual displacements to recover high-frequency surface detail such as wrinkles and contact folds that sparse bone control struggles to represent. Both stages are conditioned on a precomputed global body-garment encoding: a shape code produced by a canonical graph encoder modulates the pose-dependent network via FiLM [PSDV⁺18], while a separate decoder predicts per-vertex features from the same encoding. These features are computed once and projected onto the UV atlas, providing the UV-space network with static garment-specific priors (seam locations, panel layout and per-region stiffness) that persist across all frames. The global encoding is defined in a canonical space, disentangled from pose-dependent information, and computed once per identity.

While bone position history provides a useful temporal signal, it is insufficient as standalone training supervision because it produces degenerate velocity estimates whenever the body is stationary. This makes physics losses such as inertia uninformative. We therefore jointly train a per-vertex neural simulator based on Mesh-GraphNet [PFSG21]. Specifically, we use Hood [GBH23], which maintains explicit integration state across frames and provides physically grounded position targets via a cross-branch consistency loss. This temporal integration branch also propagates physics gradients through a shared encoder into the precomputed canonical representation, so that the cached body-garment features reflect dynamic cloth behavior rather than geometry alone. The temporal integration branch is discarded at inference; per-frame cost reduces to evaluating the compact skeleton-driven network conditioned on cached identity-specific quantities. The proposed framework achieves up to significant speedup over state-of-the-art methods while maintaining comparable reconstruction quality.

In summary, our contributions are:

- A lightweight framework that decomposes computationally expensive mesh-based integration into a compact set of bone-driven transforms, followed by an efficient convolutional

map. This design enables real-time cloth simulation on the GPU while achieving simulation quality comparable to state-of-the-art methods.

- A modulation-based global conditioning approach that decouples pose-dependent computation from the body-garment encoding thereby substantially reducing run-time cost.
- A self-supervision strategy in which a lightweight integrator model is jointly trained with a computationally intensive neural simulator. This approach enables physics-aware learning and improves the fidelity and stability of the lightweight model.

2 Related Works

Generating realistic garment deformation under tight compute constraints remains a significant challenge. While physics-based simulation offers high accuracy, its computational cost and manual setup limit scalability. Current research focuses on developing efficient, generalizable models, yet achieving a balance between performance and versatility remains an open problem.

2.1 Physics-based Simulation

Physics-based methods to simulate highly-detailed clothing is still an active research topic decades after the foundational works [BW98, Stu22]. While physics-based methods yield realistic results, their high computational cost often prohibits real-time use. Recent research addresses this by employing massively parallel algorithms to leverage GPU acceleration for interactive applications [MMC16, CLYY24]. Alternatively, simulation has been accelerated by modeling a subspace from which to solve the equations of motion [HTC⁺14, DASTH10, MSP⁺24, FMD⁺19, CCW⁺23] or predicting latent-space solutions with a neural simulator [LLL⁺25].

2.2 Skinning-based Garment Deformation

Alternative to expensive simulation methods, skinning-based garment deformation methods are a common feature in compute-limited environments like video games [JDKL14]. However, standard linear blend skinning (LBS) can create unappealing garment deformations, as the garment geometry may not map well to the underlying body structure, a problem particularly pronounced with loose-fitting garments. To improve deformation quality over traditional skinning weight methods, one can optimize skinning weights with geometric or physics-based losses and collision penalties over the skinning-based clothing [JBPS11, LSC⁺26, CCP24].

To further address improved deformations, skinning decomposition [LD12] extracts the linear blend skinning from a set of examples, effectively modeling the deformation by a low number of rigid bones. Alternatively, learning based methods employ bone-based representation to predict the deformation of loose-fitting garment meshes at interactive rates [PMJ⁺22, HSX⁺22, LTY⁺24]. More broadly, the concept of virtual bones has been widely adopted to parameterize motion across both graphics and vision. Early deformation and rigging methods [SZGP07, JBPS11, BP07] demonstrated that sparse local nodes effectively capture non-rigid deformations. This principle was later extended to real-time surface tracking in

DynamicFusion [NFS15] and LiveCap [HXZ⁺19]. Recent Gaussian-based representations, including Dynamic 3D Gaussians [LYL⁺23], RigGS [LCG⁺24], and Dual Gaussian [JSH⁺24], further parameterize complex motions through spatially distributed virtual bones. At the generative level, DiMO [MLW⁺25] distills node-based motion from diffusion models. Collectively, these works demonstrate that virtual bones form a powerful abstraction for parameterizing motion in a variety of representations, including garment deformation.

2.3 Neural Garment Simulation

In recent years, two primary approaches have emerged for learning garment simulation: supervised models relying directly on data, and self-supervised models relying on physics-based losses instead of data. Supervised methods like [HDDN19] combine subspace simulation techniques with machine learning to enable efficient subspace-only physics simulation. TailorNet [PLPM20] predicts deformations in frequency space and models clothing dynamics conditioned on pose, shape, and style. Others [LWK⁺24] leverage a manifold-aware transformer framework to predict deformations. However, these methods require simulation data that is often difficult and expensive to obtain. In contrast, self-supervised approaches have shown effectiveness in modeling physical systems without the need for precomputed training data [SOC22, BME22, SLL⁺25, LLC⁺26]. SNUG [SOC22] introduced a self-supervised training method for modeling dynamic clothing deformations in a data-free approach. NCS [BME22] extended this work by devising an architecture able to automatically disentangle static and dynamic cloth subspaces, improving quality of the results. DrapeNet [DLLG⁺23] presented a strategy enabling it to generalize across garments for modeling quasi-static deformations. HOOD [GBH23] presented a graph neural network approach to model cloth dynamics that generalizes across body and garment shapes for varying poses. Follow-up work improved performance [LSL⁺24]. Graph neural network have also been successfully employed to enable cloth upsampling [HLB⁺23, ZL24].

3 Methodology

One key observation is that garment deformation decomposes naturally by spatial frequency: global motion is well-approximated by sparse affine transforms, while surface detail such as wrinkles remains local. We exploit this through a coarse-to-fine architecture in which virtual bones capture low-frequency pose-driven motion and a UV-space network recovers high-frequency residuals. Furthermore, at runtime the mesh topology and body-garment relationship are fixed, only skeleton joint angles change per frame. We leverage this by separating identity-specific quantities, precomputed once, from pose-dependent quantities evaluated per frame by a lightweight network. To train this efficient pipeline without simulation data, we jointly optimize it alongside a per-vertex Mesh-GraphNet [PFSGB21] that integrates cloth dynamics over time under self-supervised physics losses. This temporal branch acts as a teacher, supervising the fast runtime module via a cross-branch consistency loss. Crucially, the graph encoder weights are shared between the two branches, so that physics gradients propagate into the precomputed identity representation, encoding not just canonical geometry but also the dynamic behavior it implies. The temporal

integration branch is discarded after training; at inference, only the bone-driven network runs, conditioned on cached identity-specific quantities.

The proposed framework has three modules (Figure 2): **[A]** Pose-Conditioned Deformation, the per-frame runtime branch; **[B]** Body and Garment Encoding, which computes identity-specific quantities at canonical-encoding; and **[C]** Temporal Integration, a mesh-based dynamics branch used only during training. Module [C] is discarded after training. Module [B] runs once per *identity* (body shape/garment). Only Module [A] is evaluated at run time.

3.1 Problem Setup and Virtual Bone Representation

A clothed character at time t consists of a body mesh with vertices $\mathcal{V}_b$, a skeleton of K joints with transformations $\{\mathbf{T}_k^t\}_{k=1}^K \subset \text{SE}(3)$, and a garment mesh with vertices $\mathcal{V}_c$ and fixed topology $\mathcal{E}_c$. The garment topology and UV atlas are identity properties; only skeleton joint angles, θ^t , change per frame.

Driving garment vertices directly from skeleton joints is overly restrictive since body joints encode rigid segment motion and are shared across garments, producing deformations that do not respect garment-specific behavior such as hem lag or panel-boundary compliance. Following Pan et al. [PMJ⁺22], we introduce B virtual bones as an intermediate layer: skeleton joints drive virtual bones, and virtual bones drive garment vertices. This separation allows the virtual bone layer to capture garment-local deformation modes independently of body-segment boundaries.

Virtual bone positions $\{\mathbf{p}_b^c\}_{b=1}^B$ are obtained by running Farthest Point Sampling (FPS) [ELPZ97] on the vertices of the canonically draped garment mesh. Sampling on the garment rather than the body ensures the control points are distributed according to garment surface area, not body topology, and respect panel boundaries and drape contours of the specific garment.

Skeleton $\rightarrow$ Bones. Following HOOD [GBH23], we assign each virtual bone b the SMPL skinning weights [LMR⁺15] of its nearest point on the body surface, giving per-bone weights $\mathbf{w}_b^{\text{SMPL}} \in \mathbb{R}^K$. The virtual bone position in homogeneous coordinates at time t is:

$$\hat{\mathbf{p}}_b^t = \sum_{k=1}^K \mathbf{w}_{bk}^{\text{SMPL}} \mathbf{T}_k^t \mathbf{p}_b^c. \quad (1)$$

Bones $\rightarrow$ Garment. Garment vertex blend weights $\mathbf{W}^{(0)} \in \mathbb{R}^{|\mathcal{V}_c| \times B}$ are initialized on the draped garment mesh:

$$\mathbf{w}_{ob}^{(0)} = \frac{\exp(-d_{ob}/\sigma)}{\sum_{b'} \exp(-d_{ob'}/\sigma)}, \quad (2)$$

where d_{ob} is the geodesic distance from garment vertex v to virtual bone b along the garment surface. Geodesic distances on the garment surface are used rather than Euclidean distances to avoid coupling vertices across opposite sides of a thin garment panel through the body volume [JDKL14]. These weights encode garment-surface proximity but not garment mechanics; Module [B] predicts corrections $\Delta \mathbf{w}_v$ to account for garment structure and body-shape-dependent fit.

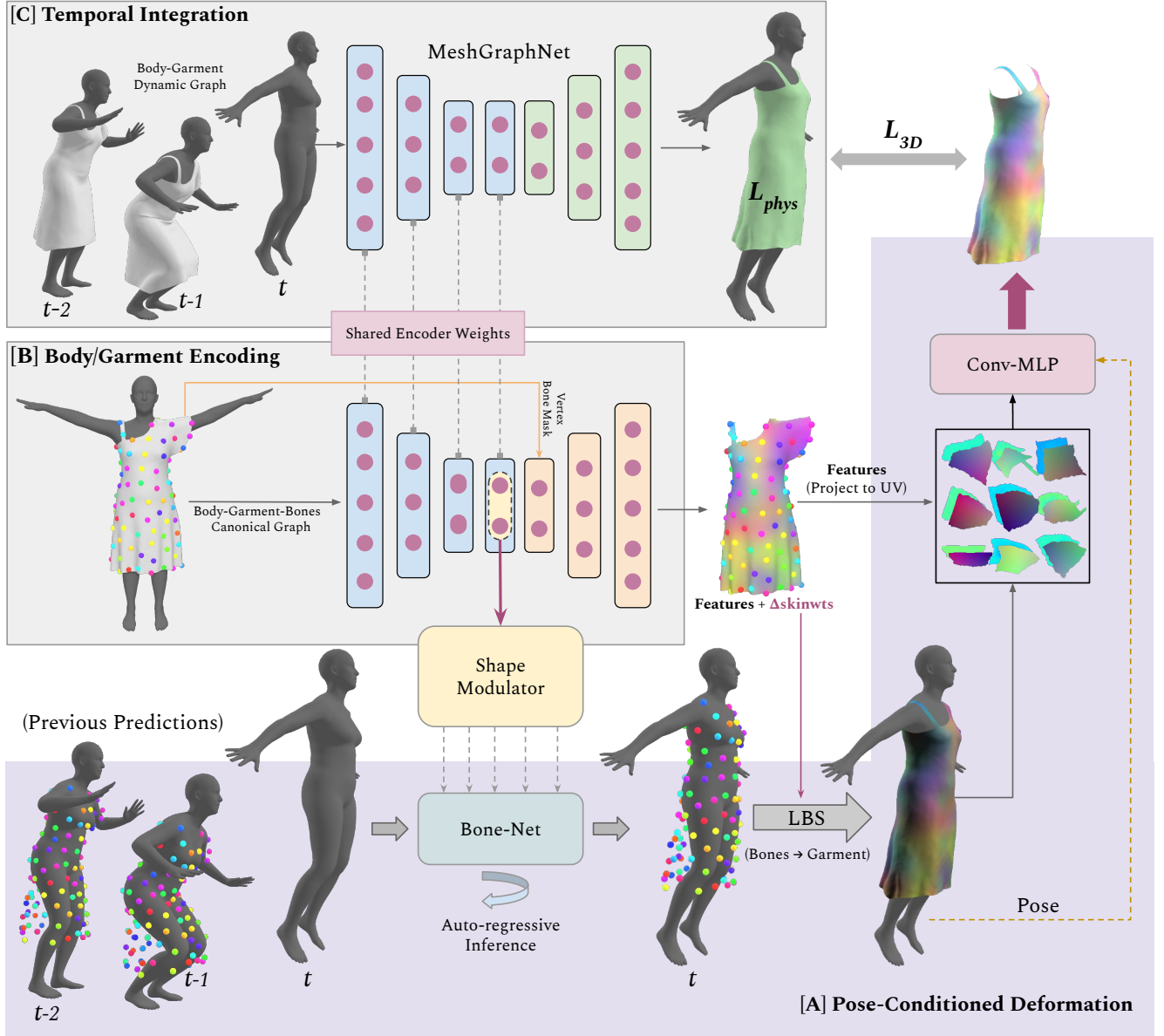


Fig. 2. **Method overview.** **[A] Pose-Conditioned Deformation:** given a pose sequence, virtual bones are first transformed via LBS (Skeleton $\rightarrow$ Bones) using SMPL skinning weights, then corrected by Bone-Net conditioned on the shape code z via the Shape Modulator. The corrected bones drive garment vertices via LBS (Bones $\rightarrow$ Garment) with learned skin weight corrections Δw . The resulting posed position map is concatenated with a static feature map and processed by a Conv-MLP to predict per-vertex residuals δp_v . **[B] Body/Garment Encoding:** a canonical Body-Garment-Bones Graph is processed once per identity through a graph encoder (weights shared with [C]) to produce z , Δw , and UV-projected features F_{UV} . **[C] Temporal Integration:** a MeshGraphNet trained on body-garment dynamic graph sequences via a self-supervised physics loss $\mathcal{L}_{phys}$; its output positions supervise Module [A] through the cross-branch consistency loss $\mathcal{L}_{3D}$. Module [C] is discarded at inference; Module [B] runs once per identity; only Module [A] runs per frame.

3.2 Module A: Pose-Conditioned Deformation

Module [A] operates at inference time and maps skeleton pose to garment vertex positions in two sequential stages, following the coarse-to-fine structure of Pan et al. [PMJ⁺22]: a global virtual-bone stage handles low-frequency pose-driven deformation, and

a UV-space stage handles high-frequency surface detail. The pose sequence $\Theta = \{\theta^{t-2}, \theta^{t-1}, \theta^t\}$ enters the pipeline via LBS (Skeleton $\rightarrow$ Bones); Bone-Net then operates on the resulting bone positions and velocities rather than on raw joint angles. Figure 3 shows the output of each stage in Module [A].

3.2.1 Stage 1: Virtual-Bone Deformation. SMPL-initialized virtual bone transforms follow body joints rigidly and carry no model of cloth dynamics. We learn per-frame corrections $\{\Delta \mathbf{T}_b^t\}_{b=1}^B$ via Bone-Net, a temporal MLP whose input at each frame is the concatenation of virtual bone positions $\{\tilde{\mathbf{p}}_b^F\}_{b=1}^B$ and bone velocities $\{\tilde{\mathbf{p}}_b^F - \tilde{\mathbf{p}}_b^{F-1}\}_{b=1}^B$ over the window $F \in \{t-2, t-1, t\}$, following the node feature convention of HOOD [GBH23]. Bone positions and their finite differences provide velocity signals directly in the garment’s reference frame, which is a stronger conditioning signal than raw joint-angle differences in local rotation space. The three-frame window provides first- and second-order temporal information sufficient to model lagged garment effects without explicit integration [SOC22, PMJ⁺22].

A single Bone-Net is shared across body shapes and garment types. Identity-specific behavior is introduced via FiLM conditioning [PSDV⁺18]: a Shape Modulator MLP maps the precomputed shape code $\mathbf{z} \in \mathbb{R}^{d_z}$ (Section 3.3) to per-layer scale and shift parameters $(\gamma_\ell, \beta_\ell)$ applied to each hidden layer $\mathbf{h}_\ell$ of Bone-Net:

$$\mathbf{h}_\ell \leftarrow \gamma_\ell(\mathbf{z}) \odot \mathbf{h}_\ell + \beta_\ell(\mathbf{z}). \quad (3)$$

Since $\mathbf{z}$ is precomputed at canonical-encoding, this adds no per-frame cost. FiLM conditioning is used here rather than concatenation or hypernetwork variants because it modulates all layers of Bone-Net from a single compact code, which has been shown to be more parameter-efficient for geometry-conditioned networks.

Module [B] predicts per-vertex corrections $\Delta \mathbf{w}_v \in \mathbb{R}^B$ to the initialized weights of Equation 2. The corrected weights are

$$\hat{\mathbf{w}}_{vb} = \text{softmax}_b \left(\mathbf{w}_{vb}^{(0)} + \Delta \mathbf{w}_{vb} \right). \quad (4)$$

The Stage-1 output position, obtained via LBS (Bones $\rightarrow$ Garment), is $\hat{\mathbf{p}}_v^t = \sum_{b=1}^B \hat{\mathbf{w}}_{vb} \left(\mathbf{T}_b^t + \Delta \mathbf{T}_b^t \right) \tilde{\mathbf{p}}_b^t$, where $\tilde{\mathbf{p}}_b^t$ is from Equation 1. Note that rotational components of transforms are added in their 6D representation before being converted to affine and applied in this equation. Weight corrections allow the model to encode garment-structure-specific influence boundaries (e.g. sharper transitions at seams, reduced dependence on proximal joints at loose hems) that the geodesic initialization cannot capture.

3.2.2 Stage 2: UV-Space Refinement. The Stage-1 output is projected onto the garment’s fixed UV atlas to produce a posed position map $\mathbf{P}_{UV}^t \in \mathbb{R}^{H \times W \times 3}$. This is concatenated with the static feature map $\mathbf{F}_{UV} \in \mathbb{R}^{H \times W \times d_f}$ from Module [B] and a broadcast pose embedding ϕ^t giving $\mathbf{X}_{UV}^t = [\mathbf{P}_{UV}^t | \mathbf{F}_{UV} | \phi^t]$. A Conv-MLP processes $\mathbf{X}_{UV}^t$ and predicts a per-vertex displacement $\delta \mathbf{p}_v$. The final garment vertex position is $\mathbf{p}_v^t = \hat{\mathbf{p}}_v^t + \delta \mathbf{p}_v$. The static feature map $\mathbf{F}_{UV}$ carries garment-specific priors (patch layout, seam locations, per-triangle stiffness) that are fixed for a given identity and condition the per-frame residual prediction without recomputation.

3.3 Module B: Body and Garment Encoding

Module [B] computes three identity-specific quantities used by Module [A]: shape code $\mathbf{z}$, skin weight corrections $\Delta \mathbf{w}$, and a static UV feature map $\mathbf{F}_{UV}$. All outputs are computed in a single forward pass over the canonical body-garment graph and precomputed.

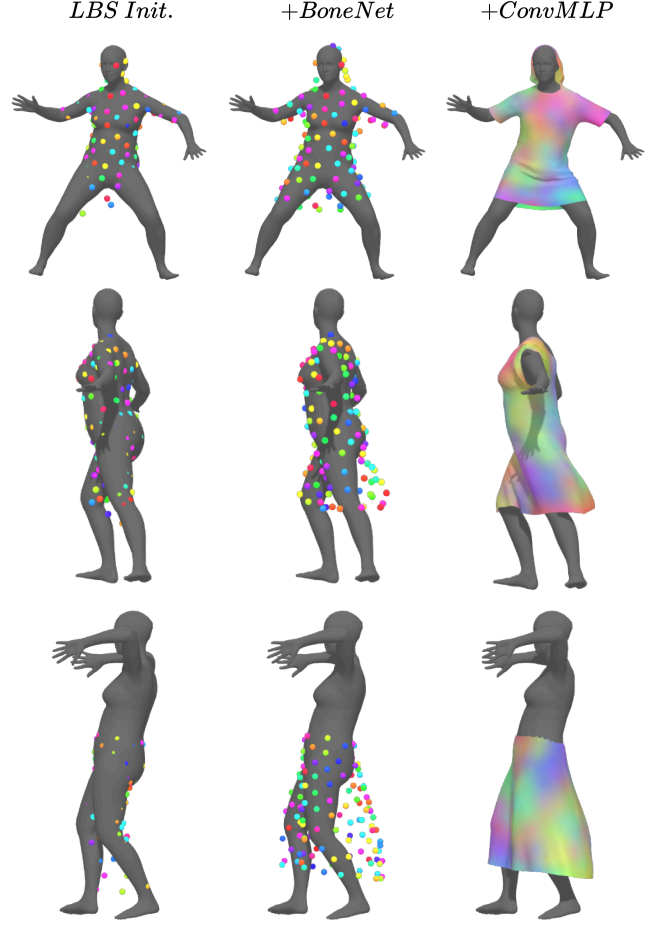


Fig. 3. Output of the individual deformation stages in Module [A] for 3 poses. During training, virtual bones (colored dots) are initially skinned using body skinning weights (left). In Stage-1, the auto-regressive coarse-level bone deformer correct bone transforms for time t (middle). After skinning the garment to the bones, we add the remaining wrinkle details in Stage-2 (right) using a Convolutional MLP.

We construct a Body-Garment-Bones Canonical Graph $\mathcal{G}_c = (\mathcal{V}_b \cup \mathcal{V}_c, \mathcal{E})$ by placing body and garment in A-pose and connecting each garment vertex to its k -nearest neighbors on the body surface.

$\mathcal{G}_c$ is processed by a message-passing graph network whose weights are shared with Module [C]. Physics gradients from Module [C]’s self-supervised loss propagate through these shared weights during training, pulling the encoder toward representations that reflect cloth-body interaction dynamics. When Module [B] passes the static canonical graph through the same encoder, the resulting per-vertex features $\mathbf{F}_c \in \mathbb{R}^{|\mathcal{V}_c| \times d_f}$ therefore carry information about dynamic behavior implied by the canonical geometry, not just the geometry itself.

Three decoder heads over $\mathbf{F}_c$ produce: (1) shape code $\mathbf{z}$, a global descriptor of the body-garment pair used to condition Bone-Net via Equation 3; (2) skin weight corrections $\Delta \mathbf{w}_v \in \mathbb{R}^B$ per garment

vertex, refining the geodesic initialization of Equation 2; and (3) per-vertex features $\mathbf{f}_v \in \mathbb{R}^{d_f}$, projected to the UV atlas to form $\mathbf{F}_{UV}$ for Stage 2.

3.4 Module C: Temporal Integration

Module [C] is a temporal integration branch trained jointly with Modules [A] and [B] and discarded after training. It serves two purposes: generating physically grounded position targets for Module [A], and providing physics gradients that train the shared encoder. Module [C] follows the HOOD architecture [GBH23]: a Mesh-GraphNet that takes a sequence of body-garment dynamic graphs $\{\mathcal{M}^{t-2}, \mathcal{M}^{t-1}, \mathcal{M}^t\}$ and predicts per-vertex cloth acceleration $\ddot{\mathbf{p}}_v$, integrated via symplectic Euler:

$$\mathbf{v}_v^t = \mathbf{v}_v^{t-1} + \Delta t \cdot \ddot{\mathbf{p}}_v^t, \quad \mathbf{p}_v^{[C],t} = \mathbf{p}_v^{t-1} + \Delta t \cdot \mathbf{v}_v^t. \quad (5)$$

3.5 Training and Inference

Training. Training proceeds in two phases. In the warm-up phase (500 epochs), only Module [C] is trained using a self-supervised physics loss ($\mathcal{L}_{\text{phys}}$) on its own integrated positions, calculated as:

$$\mathcal{L}_{\text{phys}} = \lambda_s \mathcal{L}_{\text{stretch}} + \lambda_b \mathcal{L}_{\text{bend}} + \lambda_c \mathcal{L}_{\text{collision}} + \lambda_i \mathcal{L}_{\text{inertia}} + \lambda_f \mathcal{L}_{\text{friction}} \quad (6)$$

where the inertia term is:

$$\mathcal{L}_{\text{inertia}} = \frac{1}{|\mathcal{V}_c|} \sum_v \left\| \mathbf{p}_v^{[C],t} - \left(2\mathbf{p}_v^{[C],t-1} - \mathbf{p}_v^{[C],t-2} \right) \right\|^2. \quad (7)$$

The remaining loss terms follow the implementation described in [GBH23].

The warm-up phase is necessary for Module [C] to produce physically meaningful positions before it can supervise Module [A]; at initialization, Module [C]’s outputs are arbitrary. The warmup is short relative to total training (0.5% of 100k epochs) and is sufficient to bring Module [C] into a stable physical regime. In the joint training phase (remaining 99.5k epochs), all modules are trained with

$$\mathcal{L} = \mathcal{L}_{\text{phys}} + \mathcal{L}_{3D} \quad (8)$$

where $\mathcal{L}_{\text{phys}}$ continues to train Module [C] and $\mathcal{L}_{3D}$ trains Module [A]. Module [C] and Module [A] process the same pose sequence. We supervise Module [A] to match Module [C]’s physically integrated positions via,

$$\mathcal{L}_{3D} = \mathcal{L}_{\text{MSE}} + \mathcal{L}_{\text{lap}} + \mathcal{L}_{\text{interp}} \quad (9)$$

The vertex MSE term is,

$$\mathcal{L}_{\text{MSE}} = \frac{1}{|\mathcal{V}_c|} \sum_v \left\| \mathbf{p}_v^t - \mathbf{p}_v^{[C],t} \right\|_2^2, \quad (10)$$

which transfers dynamic information, including momentum and inertial lag, into Module [A] without requiring it to compute garment velocities. Since Module [C] is jointly trained, the quality of these targets improves over the course of training. The Laplacian term,

$$\mathcal{L}_{\text{lap}} = \frac{1}{|\mathcal{V}_c|} \sum_v \left\| \mathbf{p}_v^t - \frac{1}{|\mathcal{N}(v)|} \sum_{u \in \mathcal{N}(v)} \mathbf{p}_u^t \right\|_2^2 \quad (11)$$

penalizes local surface irregularity. This is particularly relevant here because the influence boundaries of sparse virtual bones can produce sharp per-vertex discontinuities in the skinned mesh [PMJ⁺22];

the Laplacian term suppresses these without penalizing smooth macroscopic deformation. The interpenetration term,

$$\mathcal{L}_{\text{interp}} = \frac{1}{|\mathcal{V}_c|} \sum_v \max(0, \epsilon - d_v)^2 \quad (12)$$

penalizes garment vertices inside the body surface, where d_v is the signed distance to the body and ϵ is a clearance margin. This is applied directly to Module [A]’s output and is distinct from $\mathcal{L}_{\text{collision}}$ in Equation 6, which acts on Module [C]’s simulation state as an energy term; $\mathcal{L}_{\text{interp}}$ enforces geometric non-penetration on the runtime branch independently.

Gradients from $\mathcal{L}_{\text{phys}}$ flow through Module [C]’s encoder, which is shared with Module [B]. This causes Module [B]’s canonical-graph features $\mathbf{F}_c$ to reflect cloth dynamics implied by the canonical geometry, giving the decoded $\Delta \mathbf{w}_v$ and $\mathbf{f}_v$ structure that a purely geometric encoder would not produce.

Why $\mathcal{L}_{\text{phys}}$ cannot be applied directly to Module [A]? Applying $\mathcal{L}_{\text{phys}}$ directly to Module [A]’s output fails at the inertia term, which requires physically consistent garment velocities. Under virtual-bone skinning, garment vertex positions are a stateless function of the current bone positions, which are themselves determined by joint angles: $\dot{\mathbf{p}}_v^t = \mathbf{f}_v(\dot{\boldsymbol{\theta}}^t) = \mathbf{g}_v(\boldsymbol{\theta}^t)$. This has two consequences that make $\mathcal{L}_{\text{inertia}}$ uninformative when applied to skinning outputs. First, all $|\mathcal{V}_c|$ cloth velocities are a low-rank blend of the K -dimensional joint velocity, which cannot represent the locally independent velocity field of real cloth [SOC22]. Second, if $\boldsymbol{\theta}^t = \boldsymbol{\theta}^{t-1}$, then the apparent velocity $\dot{\mathbf{p}}_v^t = \mathbf{g}_v(\boldsymbol{\theta}^t) - \mathbf{g}_v(\boldsymbol{\theta}^{t-1}) = 0$ for all v regardless of prior motion, causing $\mathcal{L}_{\text{inertia}}$ to produce near-zero gradients in exactly the regime where cloth dynamics are most pronounced: post-deceleration oscillation, fabric settling, and damped swinging. Module [C] avoids this by maintaining $\mathbf{v}_v^t$ as a persistent integration variable that carries the full history of accelerations forward across frames, independent of whether the body is moving. The inertia loss on Module [C]’s integrated positions is therefore a well-posed objective throughout training. See subsection 4.4 for a visual example showing this phenomenon.

Inference. Module [C] is discarded after training. Module [B] is run once per identity to compute and cache $\mathbf{z}$, $\{\Delta \mathbf{w}_v\}$, and $\mathbf{F}_{UV}$; this cost is amortized over all subsequent sequences for that identity. Per-frame inference runs only Module [A]: LBS (Skeleton $\rightarrow$ Bones) $\rightarrow$ Bone-Net $\rightarrow$ LBS (Bones $\rightarrow$ Garment) $\rightarrow$ UV rasterization $\rightarrow$ Conv-MLP, conditioned on precomputed Module [B] outputs. The per-frame input reduces to $\boldsymbol{\Theta}$: three frames of joint angles. Per-frame cost is independent of garment mesh resolution, simulation history, and body-garment graph complexity.

4 Results & Experiments

4.1 Dataset & Experimental Setup

Following the training data strategy from GAPS[CCP24], we employ self-supervision using pose sequences from the AMASS dataset, without requiring any precomputed physics simulation data. For each garment, we evaluate on the same set of six garments from the VTO dataset: Tshirt, Tshirt-Unzipped, Pants, Long-Skirt, Dress, and Hooded-Tight-Dress. Body shapes are sampled from the SMPL model to cover a diverse range of shapes and sizes. All methods

	Tshirt			Tshirt-Unzipped			Pants			Long-Skirt			Dress			Hooded-Tight-Dress		
	ϵ_e	ϵ_a	ϵ_c	ϵ_e	ϵ_a	ϵ_c	ϵ_e	ϵ_a	ϵ_c	ϵ_e	ϵ_a	ϵ_c	ϵ_e	ϵ_a	ϵ_c	ϵ_e	ϵ_a	ϵ_c
SNUG	7.923	14.102	5.814 (± 2.391)	7.541	13.287	5.203 (± 2.187)	7.108	12.341	4.731 (± 2.043)	8.342	14.876	6.103 (± 2.812)	5.012	6.341	3.287 (± 1.623)	8.712	15.421	6.387 (± 3.012)
NCS	3.782	5.498	0.587 (± 0.541)	4.103	5.812	0.652 (± 0.601)	4.521	5.843	2.187 (± 0.712)	4.287	6.123	0.801 (± 0.658)	4.312	5.623	0.412 (± 0.589)	4.923	6.912	0.698 (± 1.087)
GAPS	3.487	5.431	0.132 (± 0.443)	3.187	3.971	0.214 (± 0.321)	2.456	3.143	0.031 (± 0.148)	3.298	4.187	0.158 (± 0.334)	3.187	3.971	0.213 (± 0.327)	3.812	4.423	0.141 (± 0.243)
HOOD	3.102	4.876	0.098 (± 0.381)	2.934	3.701	0.172 (± 0.289)	1.721	2.412	0.009 (± 0.058)	2.981	3.876	0.131 (± 0.297)	2.897	3.654	0.181 (± 0.291)	2.712	3.521	0.058 (± 0.114)
Ours w/o $\mathcal{L}_{lap}$	3.712	5.698	0.161 (± 0.478)	3.341	4.187	0.247 (± 0.358)	2.634	3.312	0.043 (± 0.172)	3.487	4.423	0.187 (± 0.361)	3.354	4.198	0.241 (± 0.354)	4.023	4.687	0.168 (± 0.271)
Ours w/o $\mathcal{L}_{interp}$	3.412	5.187	6.134 (± 2.587)	3.198	4.012	5.498 (± 2.312)	2.143	2.834	5.043 (± 2.187)	3.287	4.198	6.412 (± 2.934)	3.198	3.987	3.512 (± 1.743)	3.412	4.187	6.712 (± 3.143)
Ours w/o Conv-Net MLP	4.312	6.187	0.823 (± 0.612)	4.587	6.412	0.912 (± 0.687)	4.921	6.298	2.512 (± 0.823)	4.812	6.798	1.187 (± 0.798)	4.612	5.987	0.687 (± 0.654)	5.312	7.487	1.012 (± 1.187)
OURS	2.541	3.987	0.063 (± 0.241)	2.312	3.101	0.118 (± 0.187)	1.698	2.387	0.012 (± 0.071)	2.398	3.187	0.089 (± 0.198)	2.287	3.076	0.121 (± 0.193)	2.687	3.487	0.074 (± 0.128)

Table 1. Quantitative comparison across garments (green is best, Orange is second best)

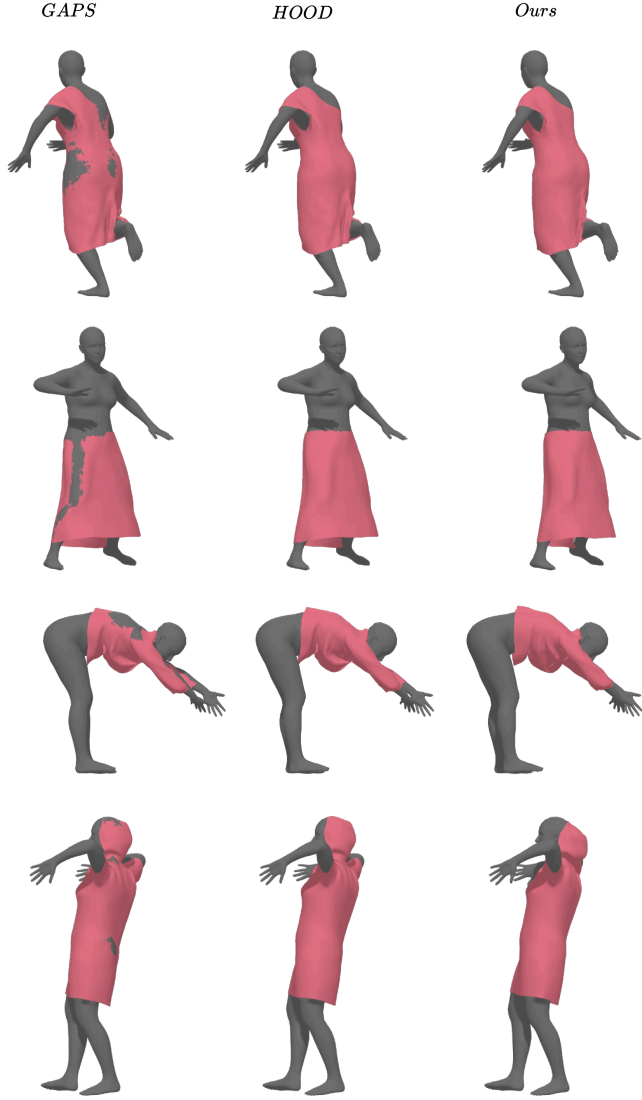


Fig. 4. **Comparison:** GAPS relies on a GRU-based motion state that accumulates drift over long sequences, leading to progressive deformation artifacts. Our method adopts HOOD’s autoregressive integration scheme, maintaining stable simulation quality regardless of sequence length but with much improved performance (see supplementary video).

Method	Tshirt	Tshirt-Unzipped	Pants	Long-Skirt	Dress	Hooded-Tight-Dress
GAPS	2.02	2.02	1.98	2.72	3.02	2.14
HOOD	27.45	26.72	25.22	36.67	49.52	36.61
OURS	1.07	1.07	1.05	1.16	1.31	1.14

Table 2. Inference time (ms) per garment; Ours achieve highest simulation FPS while maintaining superior quality.

	Tshirt-Unzipped			Dress		
No. of Bones	ϵ_e	ϵ_a	ϵ_c	ϵ_e	ϵ_a	ϵ_c
32	4.812	6.412	0.687 (± 0.598)	8.234	11.876	4.123 (± 2.187)
64	2.687	3.487	0.213 (± 0.254)	5.432	7.298	2.187 (± 1.412)
128	2.312	3.101	0.118 (± 0.187)	2.287	3.076	0.121 (± 0.193)

Table 3. Effect of number of bones on the Tshirt-Unzipped and Dress

are evaluated on a held-out mixture of AMASS and VTO sequences that are not used during training. For quantitative assessment, we adopt the error metrics introduced in [CCP24]. Specifically, Edge Error ϵ_e measures the mean percentage deviation in edge lengths between the rest-pose template and the draped garment, while Area Error (ϵ_a) quantifies the mean percentage deviation in triangle face areas between the template and the draped garment. We refer the reader to Chen et al. (GAPS) [CCP24] for the complete mathematical definitions of these metrics. In addition, ϵ_c denotes the percentage of garment vertices that penetrate the body surface, providing a direct measure of physical plausibility; all values are reported prior to any post-processing correction.

4.2 Generalization to Different Body Shapes

We evaluate generalization to unseen body shapes by testing on a diverse set of sampled SMPL body shape parameters not seen during training. As shown in Figure 5, our method produces physically plausible garment draping across all body shapes, demonstrating that the FiLM-conditioned shape modulation in Bone-Net effectively decouples identity-specific garment behavior from per-frame pose dynamics.

4.3 Comparison with SOTA methods.

We compare our approach against SNUG [SOC22], NCS [BME22], GAPS [CCP24], and HOOD [GBH23] on six garment types that span a broad range of topologies. As shown in Table 1, our method attains the lowest values of ϵ_e and ϵ_a across all garments, with the sole exception of ϵ_c on tight-fitting garments (Pants, Hooded-Tight-Dress), where HOOD’s specialized graph-based collision handling results in slightly reduced penetration. GAPS exhibits cumulative drift over long sequences due to its GRU-based motion state representation, whereas HOOD yields temporally stable simulations at a substantially higher computational cost. In contrast, our method delivers

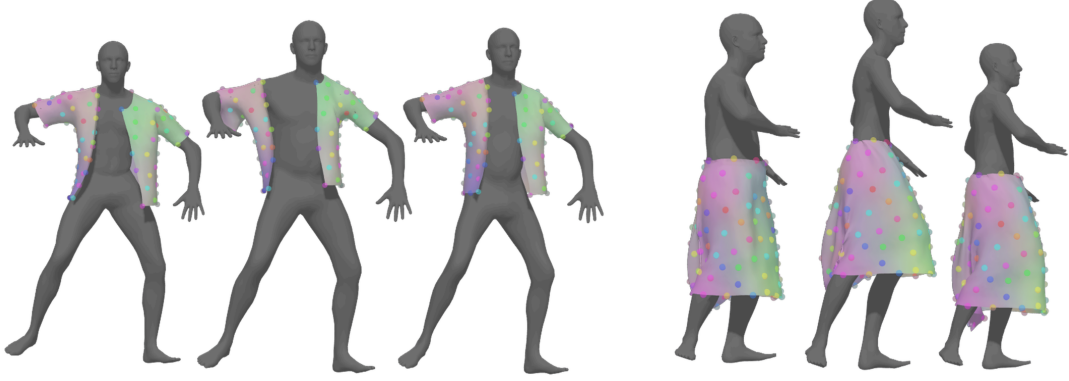


Fig. 5. Generalization to multiple body shapes.

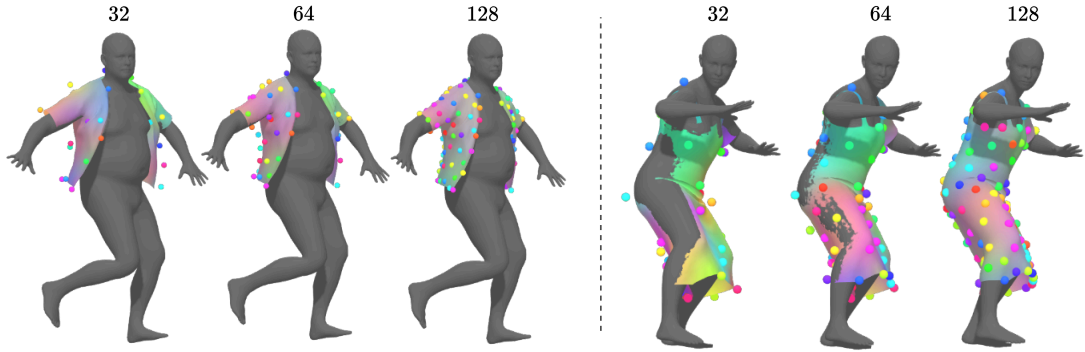


Fig. 6. **Ablation on Number of Bones:** For most garments, 32 bones lead to interpenetration and 64 yield acceptable but imperfect results. For most garments, fewer than 128 bones produce visually degraded deformations (see supplementary video).

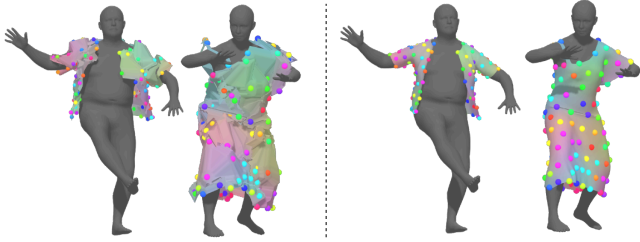


Fig. 7. **Ablation over Temporal Integration:** Applying the physics loss directly to Module A’s output produces unstable dynamics (left), as sparse bone transforms cannot represent dense per-vertex inertial states. Supervising Module A through the per-vertex GNN oracle (Module C) yields stable, physically plausible results (right).

comparable or superior simulation accuracy while operating at more than $25\times$ the inference speed of HOOD.

Figure 4 shows qualitative comparison with the two best-performing baseline methods, HOOD and GAPS. As shown, GAPS suffers from pronounced misalignment, which we attribute to its pre-skinning delta updates and its GRU-based pose history representation, both of which tend to become noisy over long motion sequences. In

contrast, the autoregressive formulation employed by HOOD, and likewise by our method, does not exhibit such limitations.

4.4 Ablation Study

We conduct an ablation study over several architectural design choices and loss components in our pipeline, as summarized in Table 1. Additionally, we provide an argument for the necessity of temporal alignment module and an analysis over number of bones below.

Local Refinement via ConvMLP: Results from Table 1 highlights the impact of the ConvMLP module: its removal leads to severe interpenetration artifacts and degenerated triangles (large area error), which we attribute to the limited expressiveness of a purely bone-based representation (even when equipped with nonlinear corrections). Since the underlying dynamics are predominantly local rather than global, the inclusion of a dedicated local refinement branch—implemented as the ConvMLP in our case, is essential for accurately modeling these effects.

Regularization terms in 3D Loss: We ablate the regularization terms $\mathcal{L}_{\text{lap}}$ and $\mathcal{L}_{\text{interp}}$ in the $\mathcal{L}_{3d}$ loss between Module [C] and Module [A]. When $\mathcal{L}_{\text{lap}}$ is removed, the resulting meshes exhibit larger error margins and degenerate triangles, caused by the lack of any global alignment between the output of Module [A] and the

output of Module [C]. Therefore, the Laplacian regularizer $\mathcal{L}_{\text{lap}}$ is crucial for enforcing global geometric consistency and preventing degeneracies.

Temporal Alignment vs. Direct Physics-based Supervision: Figure 7 presents an evaluation of our MeshGraphNet-based temporal alignment strategy (Module [C]) against the alternative approach of directly imposing physics-based supervision on Module [A]. Although the bone representations can, in principle, learn the underlying simulation dynamics over time, we observe that enforcing this supervision directly on Module [A] frequently leads to an abrupt breakdown of the simulation in early epochs. This failure mode is predominantly induced by early, skinning-induced interpenetrations, which produces degenerate simulation geometry, creating simulation instability. For this reason, we first train Module [C] in a warm-up phase for 500 epochs, thereby ensuring that the bone-deformation module receives physically plausible gradient updates.

Number of Bones: The number of bones is a crucial factor, as it affects both the computational complexity and the ability of a network with limited linear behavior to learn over an inherently non-linear domain. We aim to minimize the number of bones to keep BoneNet’s input dimensionality sufficiently low for real-time operation, while simultaneously preserving its capacity to approximate complex simulation spaces. In order to determine the ideal number of bones, we conduct an ablation study on two types of garments: a relatively simple, small garment (Tshirt-Unzipped) and a loose garment exhibiting more free-form deformation. Table 3 and Figure 6 demonstrate quantitatively and qualitatively that 128 bones provide sufficient representational power to capture complex non-linear simulation dynamics. Additionally, Table 2 indicates that our method remains significantly faster than previous works.

5 Limitations and Future Work

We demonstrate that our proposed architecture generalizes over body shape and pose for a set of six various garments. However, full generalizability is desirable for unlocking a completely automated pipeline where newly generated garments can be animated immediately. Recent advances in generative techniques have dramatically improved garment creation, allowing users to design new clothing from simple text prompts, scans, or images [CLK⁺25, SMR⁺25, LCL⁺24, SSX⁺25]. Enabling garment deformation without the need for garment specific models will unlock numerous applications.

6 Conclusion

We introduced a novel and efficient method to simulate worn garment dynamics across diverse body shapes and poses. Our approach proposes a novel architecture for distilling large and computationally expensive neural integration into a lightweight system achieving around 25x performance boost on GPU. The system generalizes to arbitrary body shapes and motion, and can support up to six garments in a single trained model. Through comprehensive experiments, we show that our method consistently outperforms state-of-the-art approaches, enabling garment simulation for real-time applications.

References

- [BME22] Hugo Bertiche, Meysam Madadi, and Sergio Escalera. Neural cloth simulation. *ACM Transactions on Graphics (TOG)*, 41(6):1–14, 2022.
- [BP07] Ilya Baran and Jovan Popović. Automatic rigging and animation of 3d characters. In *ACM SIGGRAPH*, 2007.
- [BW98] David Baraff and Andrew Witkin. Large steps in cloth simulation. In *Proceedings of the 25th Annual Conference on Computer Graphics and Interactive Techniques, SIGGRAPH '98*, page 43–54, New York, NY, USA, 1998. Association for Computing Machinery.
- [CCP24] Ruochen Chen, Liming Chen, and Shaifali Parashar. Gaps: Geometry-aware, physics-based, self-supervised neural garment draping. In *2024 International Conference on 3D Vision (3DV)*, pages 116–125. IEEE, 2024.
- [CCW⁺23] Yue Chang, Peter Yichen Chen, Zhecheng Wang, Maurizio M. Chiaramonte, Kevin Carlberg, and Eitan Grinspun. Licrom: Linear-subspace continuous reduced order modeling with neural fields. In *SIGGRAPH Asia 2023 Conference Papers*, SA '23, New York, NY, USA, 2023. Association for Computing Machinery.
- [CLK⁺25] Hsiao-Yu Chen, Egor Larionov, Ladislav Kavan, Gene Lin, Doug Roble, Olga Sorkine-Hornung, and Tuur Stuyck. Dress anyone: Automatic physically-based garment pattern refitting. *Proc. ACM Comput. Graph. Interact. Tech.*, 8(4), August 2025.
- [CLY24] Anka He Chen, Ziheng Liu, Yin Yang, and Cem Yuksel. Vertex block descent. *ACM Transactions on Graphics (TOG)*, 43(4):1–16, 2024.
- [DASTH10] Edilson De Aguiar, Leonid Sigal, Adrien Treuille, and Jessica K Hodgins. Stable spaces for real-time clothing. *ACM Transactions on Graphics (TOG)*, 29(4):1–9, 2010.
- [DLLG⁺23] Luca De Luigi, Ren Li, Benoît Guillard, Mathieu Salzmann, and Pascal Fua. Drapenet: Garment generation and self-supervised draping. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1451–1460, 2023.
- [ELPZ97] Yuval Eldar, Michael Lindenbaum, Moshe Porat, and Yehoshua Y Zeevi. The farthest point strategy for progressive image sampling. *IEEE transactions on image processing*, 6(9):1305–1315, 1997.
- [FMD⁺19] Lawson Fulton, Vismay Modi, David Duvenaud, David I. W. Levin, and Alec Jacobson. Latent-space dynamics for reduced deformable simulation. *Computer Graphics Forum*, 2019.
- [GBH23] Artur Grigorev, Michael J. Black, and Otmar Hilliges. Hood: Hierarchical graphs for generalized modelling of clothing dynamics. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16965–16974, June 2023.
- [HDDN19] Daniel Holden, Bang Chi Duong, Sayantan Datta, and Derek Nowrouzezahrai. Subspace neural physics: Fast data-driven interactive simulation. In *Proceedings of the 18th annual ACM SIGGRAPH/Eurographics Symposium on Computer Animation*, pages 1–12, 2019.
- [HLB⁺23] Oshri Halimi, Egor Larionov, Zohar Barzelay, Philipp Herholz, and Tuur Stuyck. Physgraph: Physics-based integration using graph neural networks. *arXiv preprint arXiv:2301.11841*, 2023.
- [HSX⁺22] Oshri Halimi, Tuur Stuyck, Donglai Xiang, Timur M Bagautdinov, He Wen, Ron Kimmel, Takaaki Shiratori, Chenglei Wu, Yaser Sheikh, and Fabian Prada. Pattern-based cloth registration and sparse-view animation. *ACM Trans. Graph.*, 41(6):196–1, 2022.
- [HTC⁺14] Fabian Hahn, Bernhard Thomaszewski, Stelian Coros, Robert W Sumner, Forrester Cole, Mark Meyer, Tony DeRose, and Markus Gross. Subspace clothing simulation using adaptive bases. *ACM Transactions on Graphics (TOG)*, 33(4):1–9, 2014.
- [HXZ⁺19] Marc Habermann, Weipeng Xu, Michael Zollhöfer, Gerard Pons-Moll, and Christian Theobalt. Livecap: Real-time human performance capture from monocular video. In *ACM Transactions on Graphics (Proc. SIGGRAPH)*, 2019.
- [JBPS11] Alec Jacobson, Ilya Baran, Jovan Popović, and Olga Sorkine. Bounded biharmonic weights for real-time deformation. In *ACM SIGGRAPH*, 2011.
- [JDKL14] Alec Jacobson, Zhigang Deng, Ladislav Kavan, and JP Lewis. Skinning: Real-time shape deformation. In *ACM SIGGRAPH 2014 Courses*, 2014.
- [JSH⁺24] Yuheng Jiang, Zhehao Shen, Yu Hong, Chengcheng Guo, Yize Wu, Yingliang Zhang, Jingyi Yu, and Lan Xu. Robust dual gaussian splatting for immersive human-centric volumetric videos. *arXiv preprint arXiv:2409.08353*, 2024.
- [LCG⁺24] Yinghao Liu, Enric Corona, Jiatao Gu, Christian Theobalt, Justus Thies, and Timo Bolkart. Riggs: Learning rigs for category-level 3d shape and pose estimation. In *CVPR*, 2024.
- [LCL⁺24] Yifei Li, Hsiao-yu Chen, Egor Larionov, Nikolaos Sarafianos, Wojciech Matusik, and Tuur Stuyck. Diffavatar: Simulation-ready garment optimization with differentiable simulation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4368–4378, June 2024.

- [LD12] Binh Huy Le and Zhigang Deng. Smooth skinning decomposition with rigid bones. *ACM Transactions on Graphics (TOG)*, 31(6):1–10, 2012.
- [LLC⁺26] Gene Wei-Chin Lin, Egor Larionov, Hsiao-yu Chen, Doug Roble, and Tuur Stuyck. Neuralocks: Real-time dynamic neural hair simulation. In *Computer Graphics Forum*, page e70407. Wiley Online Library, 2026.
- [LLL⁺25] Yue Li, Gene Wei-Chin Lin, Egor Larionov, Aljaz Bozic, Doug Roble, Ladislav Kavan, Stelian Coros, Bernhard Thomaszewski, Tuur Stuyck, and Hsiao-Yu Chen. Self-supervised learning of latent space dynamics. *Proc. ACM Comput. Graph. Interact. Tech.*, 8(4), August 2025.
- [LMR⁺15] Matthew Loper, Naureen Mahmood, Javier Romero, Gerard Pons-Moll, and Michael J. Black. SMPL: A skinned multi-person linear model. *ACM Trans. Graphics (Proc. SIGGRAPH Asia)*, 34(6):248:1–248:16, October 2015.
- [LSC⁺26] Egor Larionov, Igor Santesteban, Hsiao-yu Chen, Gene Wei-Chin Lin, Philipp Herholz, Ryan Goldade, Ladislav Kavan, Doug Roble, and Tuur Stuyck. Skincells: Sparse skinning using voronoi cells. In *Computer Graphics Forum*, page e70381. Wiley Online Library, 2026.
- [LSL⁺24] Tianxing Li, Rui Shi, Zihui Li, Takashi Kanai, and Qing Zhu. Efficient deformation learning of varied garments with a structure-preserving multilevel framework. *Proceedings of the ACM on Computer Graphics and Interactive Techniques*, 7(1):1–19, 2024.
- [LTY⁺24] Yudi Li, Min Tang, Yun Yang, Ruofeng Tong, Shuangcai Yang, Yao Li, Bailin An, and Qilong Kou. Ctsn: Predicting cloth deformation for skeleton-based characters with a two-stream skinning network. *Computational Visual Media*, 10(3):471–485, 2024.
- [LWK⁺24] Peizhuo Li, Tuanfeng Y Wang, Timur Levent Kesdogan, Duygu Ceylan, and Olga Sorkine-Hornung. Neural garment dynamics via manifold-aware transformers. In *Computer Graphics Forum*, volume 43, page e15028. Wiley Online Library, 2024.
- [LYL⁺23] Jonas Luiten, Xin Yu, Bastian Leibe, Luc Van Gool, and Andreas Geiger. Dynamic 3d gaussians: Tracking by learning an implicit deformation graph. In *NeurIPS*, 2023.
- [MLW⁺25] Linzhan Mou, Jiahui Lei, Chen Wang, Lingjie Liu, and Kostas Daniilidis. Dimo: Diverse 3d motion generation for arbitrary objects. In *ICCV*, 2025.
- [MMC16] Miles Macklin, Matthias Müller, and Nuttapon Chentanez. Xpbd: position-based simulation of compliant constrained dynamics. In *Proceedings of the 9th International Conference on Motion in Games*, pages 49–54, 2016.
- [MSP⁺24] Vismay Modi, Nicholas Sharp, Or Perel, Shinjiro Sueda, and David I. W. Levin. Simplicits: Mesh-free, geometry-agnostic elastic simulation. *ACM Trans. Graph.*, 43(4), July 2024.
- [NFS15] Richard A. Newcombe, Dieter Fox, and Steven M. Seitz. Dynamicfusion: Reconstruction and tracking of non-rigid scenes in real time. In *CVPR*, 2015.
- [PFSGB21] Tobias Pfaff, Meire Fortunato, Alvaro Sanchez-Gonzalez, and Peter Battaglia. Learning mesh-based simulation with graph networks. In *International Conference on Learning Representations*, 2021.
- [PLPM20] Chaitanya Patel, Zhouyingcheng Liao, and Gerard Pons-Moll. Tailornet: Predicting clothing in 3d as a function of human pose, shape and garment style. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 7365–7375, 2020.
- [PMJ⁺22] Xiaoyu Pan, Jiaming Mai, Xinwei Jiang, Dongxue Tang, Jingxiang Li, Tianjia Shao, Kun Zhou, Xiaogang Jin, and Dinesh Manocha. Predicting loose-fitting garment deformations using bone-driven motion networks. In *ACM SIGGRAPH 2022 Conference Proceedings*, SIGGRAPH '22, New York, NY, USA, 2022. Association for Computing Machinery.
- [PSDV⁺18] Ethan Perez, Florian Strub, Harm De Vries, Vincent Dumoulin, and Aaron Courville. Film: Visual reasoning with a general conditioning layer. In *Proceedings of the AAAI conference on artificial intelligence*, volume 32, 2018.
- [SLL⁺25] Tuur Stuyck, Gene Wei-Chin Lin, Egor Larionov, Hsiao-yu Chen, Aljaz Bozic, Nikolaos Sarafianos, and Doug Roble. Quaffure: Real-time quasi-static neural hair simulation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 239–249, 2025.
- [SMR⁺25] Astitva Srivastava, Pranav Manu, Amit Raj, Varun Jampani, and Avinash Sharma. Wordrobe: Text-guided generation of textured 3d garments. In Aleš Leonardis, Elisa Ricci, Stefan Roth, Olga Russakovsky, Torsten Sattler, and Gül Varol, editors, *Computer Vision – ECCV 2024*, pages 458–475, Cham, 2025. Springer Nature Switzerland.
- [SOC22] Igor Santesteban, Miguel A. Otaduy, and Dan Casas. Snug: Self-supervised neural dynamic garments. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8140–8150, June 2022.
- [SSX⁺25] Nikolaos Sarafianos, Tuur Stuyck, Xiaoyu Xiang, Yilei Li, Jovan Popovic, and Rakesh Ranjan. Garment3dgen: 3d garment stylization and texture generation. In *2025 International Conference on 3D Vision (3DV)*, pages 1382–1393. IEEE, 2025.
- [Stu22] Tuur Stuyck. *Cloth simulation for computer graphics*. Springer Nature, 2022.
- [SZGP07] Robert W. Sumner, Matthias Zwicker, Craig Gotsman, and Mark Pauly. Embedded deformation for shape manipulation. In *ACM SIGGRAPH*, 2007.
- [WSS⁺25] Shaofei Wang, Tomas Simon, Igor Santesteban, Timur Bagautdinov, Junxuan Li, Vasu Agrawal, Fabian Prada, Shou-I Yu, Pace Nalbone, Matt Gramlich, Roman Lubachersky, Chenglei Wu, Javier Romero, Jason Saragih, Michael Zollhoefer, Andreas Geiger, Siyu Tang, and Shunsuke Saito. Re-lightable full-body gaussian codec avatars. In *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Papers*, SIGGRAPH Conference Papers '25, New York, NY, USA, 2025. Association for Computing Machinery.
- [ZL24] Meng Zhang and Jun Li. Neural garment dynamic super-resolution. In *SIGGRAPH Asia 2024 Conference Papers*, pages 1–11, 2024.